

we are looking for $\frac{\partial f_{\text{loss}}}{\partial h_{\text{predict},i}}$

given $\frac{\partial f_{\text{loss}}}{\partial h_{\text{predict},i}}$

Let: $h_{\text{predict}} = (x_{ij})_{1 \leq i \leq m, 1 \leq j \leq n}$

$h_{\text{predict}} = (y_{ij})$

$$\hat{x}_{ij} = \frac{x_{ij} - \mu_{B_j}}{\sqrt{\sigma_{B_j}^2 + \epsilon}} \quad \left\{ \begin{array}{l} \sigma_{B_j}^2 = \frac{1}{m-1} \sum_{i=1}^m (x_{ij} - \mu_{B_j})^2 \\ \mu_{B_j} = \frac{1}{m} \sum_{i=1}^m x_{ij} \end{array} \right.$$

$$\frac{\partial f_{\text{loss}}}{\partial x_{ij}} = \sum_{i,j} \frac{\partial f_{\text{loss}}}{\partial y_{ij}} \frac{\partial y_{ij}}{\partial x_{ij}} \quad \frac{\partial f_{\text{loss}}}{\partial y_{ij}} = \sum_{i,j} \frac{\partial f_{\text{loss}}}{\partial \hat{x}_{ij}} \frac{\partial \hat{x}_{ij}}{\partial y_{ij}}$$

$$\frac{\partial f_{\text{loss}}}{\partial \hat{x}_{ij}} = \gamma \cdot \frac{\partial f_{\text{loss}}}{\partial y_{ij}}$$

$$\frac{\partial f_{\text{loss}}}{\partial x_{ij}} = \frac{\partial f_{\text{loss}}}{\partial \hat{x}_{ij}} \cdot \frac{1}{\sqrt{\sigma_{B_j}^2 + \epsilon}}$$

$$\frac{\partial f_{\text{loss}}}{\partial \sigma_{B_j}^2} = \sum_{i=1}^m - \frac{\partial f_{\text{loss}}}{\partial \hat{x}_{ij}} \frac{(x_{ij} - \mu_{B_j}) \sigma_{B_j}}{(\sigma_{B_j}^2 + \epsilon)^{3/2}}$$

$$\frac{\partial f_{\text{loss}}}{\partial \mu_{B_j}} = \sum_{i=1}^m - \frac{\partial f_{\text{loss}}}{\partial \hat{x}_{ij}} \cdot \frac{1}{\sqrt{\sigma_{B_j}^2 + \epsilon}}$$

$$\frac{\partial f_{\text{loss}}}{\partial x_{ij}} = \sum_{i,j} \frac{\partial f_{\text{loss}}}{\partial x_{ij}} \frac{\partial x_{ij}}{\partial x_{ij}} + \sum_j \frac{\partial f_{\text{loss}}}{\partial \mu_{B_j}} \frac{\partial \mu_{B_j}}{\partial x_{ij}} + \sum_j \frac{\partial f_{\text{loss}}}{\partial \sigma_{B_j}^2} \frac{\partial \sigma_{B_j}^2}{\partial x_{ij}}$$

$$\frac{\partial f_{\text{loss}}}{\partial x_{ij}} = \frac{\partial f_{\text{loss}}}{\partial x_{ij}} + \frac{\partial f_{\text{loss}}}{\partial \mu_{B_j}} \cdot \frac{1}{m} + \sum_j \frac{\partial f_{\text{loss}}}{\partial \sigma_{B_j}^2} \cdot \frac{2}{m-1} \cdot \frac{(x_{ij} - \mu_{B_j})}{\sigma_{B_j}}$$

$$\frac{\partial \text{loss}}{\partial x_{ij}} = \left[\frac{\partial \text{loss}}{\partial \hat{x}_{ij}} + \sum_{k=1}^m \frac{\partial \text{loss}}{\partial \hat{x}_{kj}} \cdot \frac{1}{m} + \sum_{k=1}^m \frac{\partial \text{loss}}{\partial x_{ij}} \cdot \frac{\nabla_{\beta_j} (x_{kj} - \mu_{\beta_j}) (x_{ij} - \mu_{\beta_j})}{(\nabla_{\beta_j}^2 + \epsilon) \nabla_{\beta_j}} \right] \frac{1}{\sqrt{\nabla_{\beta_j}^2 + \epsilon}}$$

$$\frac{\partial \text{loss}}{\partial x_{ij}} = \frac{\gamma}{m \sqrt{\nabla_{\beta_j}^2 + \epsilon}} \left[m \frac{\partial \text{loss}}{\partial y_{ij}} + \sum_{k=1}^m \frac{\partial \text{loss}}{\partial y_{kj}} - \sum_{k=1}^m \frac{\partial \text{loss}}{\partial y_{ij}} \cdot \hat{x}_{kj} \cdot \hat{x}_{ij} \cdot \frac{m}{m-1} \right]$$



$$\frac{\partial \text{loss}}{\partial x_{ij}} = \frac{\gamma}{m \sqrt{\nabla_{\beta_j}^2 + \epsilon}} \left[m \frac{\partial \text{loss}}{\partial y_{ij}} + \sum_{k=1}^m \frac{\partial \text{loss}}{\partial y_{kj}} - \sum_{k=1}^m \frac{\partial \text{loss}}{\partial y_{ij}} \cdot \hat{x}_{kj} \cdot \hat{x}_{ij} \cdot \frac{m}{m-1} \right]$$